30-Day Python Roadmap

Personalized for a working professional transitioning into Python for workplace automation & data tasks

Your Profile

Situation: Working Professional (non-tech) | Current Skills: Basic computer literacy, no coding | Goal: Automate spreadsheets/reports & build data skills for career growth | Time: 1 hour/day | Level: Beginner | Style: Projects + Videos

Final Outcome

By Day 30, you'll comfortably write Python scripts to automate repetitive work tasks (reports, spreadsheets, file organization, emails), pull data from files/APIs, and present it clearly. You'll finish with a working portfolio project: an automated report generator you can actually use at your job or show in interviews.

Week 1: Foundations Without Overwhelm

Milestone: Comfortable with Python syntax, variables, and basic logic — using tasks tied to real spreadsheet/report scenarios.

Day	Task	Resource
1	Install Python + VS Code. Write & run your first script (print, comments, variables).	Python.org downloads; VS Code Python extension
2	Learn data types (numbers, text, booleans) using an expense-tracking example.	"Python for Everybody" video, Ch.2 (Dr. Chuck, YouTube)
3	String basics: formatting messages like real report text (f-strings).	Corey Schafer – Python Strings (YouTube)
4	If/else logic: build a simple rule (e.g. flag expenses over $100).	freeCodeCamp Python course, Conditionals section
5	Loops: process a list of sample data (e.g. weekly sales numbers).	Programiz – Python for/while loops
6	Lists & basic list operations using a sample task list.	Python docs – Data Structures (skim only)
7	Weekend Project: Build a script that summarizes a list of expenses (total, average, highest).	—

Week 2: Real Data Handling

Milestone: Able to work with files and structured data — the core skill for workplace automation.

Day	Task	Resource
8	Dictionaries: model a real record (e.g. an employee or product entry).	Real Python – Dictionaries (video version if available)
9	Functions: turn Week 1's expense summary into a reusable function.	Corey Schafer – Functions (YouTube)

Day	Task	Resource
10	Reading/writing CSV files (this is the automation building block).	Real Python video – "Reading CSVs in Python"
11	Practice: load a CSV and calculate totals/averages per category.	Kaggle sample datasets (small CSVs)
12	Error handling (try/except) — make your script fail gracefully on bad data.	Tech With Tim – Error Handling (YouTube)
13	Intro to pandas (just enough to read/filter a spreadsheet).	pandas 10-minute intro video (YouTube)
14	Weekend Project: Build a script that reads a CSV report and outputs a clean summary file.	—

Week 3: Automation Skills

Milestone: Able to automate a real repetitive task end-to-end.

Day	Task	Resource
15	Working with Excel files directly using openpyxl.	openpyxl official quickstart (docs + example video)
16	Automate a folder task: rename/organize files by rule.	Corey Schafer – os module (YouTube)
17	Sending automated emails with Python (great workplace skill).	Tech With Tim – "Send Emails with Python" (YouTube)
18	Combine skills: read data → process → export to Excel.	Review Day 13–17 material
19	Intro to APIs: pull live data (e.g. currency rates, weather).	freeCodeCamp – "APIs for Beginners" (YouTube)
20	Practice: build a mini script that fetches API data and saves it to a file.	Public APIs list (GitHub: public-apis/public-apis)
21	Weekend Project: Automate a weekly report — pull data, process it, export a formatted Excel file.	—

Week 4: Portfolio Project & Wrap-Up

Milestone: Ship a complete, usable automation project and consolidate your learning.

Day	Task	Resource
22	Plan your final project: pick a real repetitive task from your job/life to automate.	—
23	Debugging basics — reading error messages calmly and fixing them.	Real Python video – "Python Debugging"
24	Version control basics: create a GitHub account, learn git add/commit/push.	GitHub's Git Handbook (beginner video walk-through)
25	Build final project — Part 1 (core logic).	—
26	Build final project — Part 2 (data handling + output).	—

Day	Task	Resource
27	Build final project — Part 3 (polish, error handling, comments).	—
28	Write a simple README explaining what your tool does.	makeareadme.com
29	Push project to GitHub; test it end-to-end.	—
30	Reflect: what you can now automate at work, and plan your next learning step (e.g. pandas deep-dive or a Flask app).	—

Final Project Options

- Automated Weekly Report Generator — reads raw data (CSV/Excel), calculates summaries, and outputs a formatted report.
- Inbox/File Organizer — sorts files or emails into folders based on rules you define.
- Expense/Budget Tracker — logs and categorizes expenses, generates a monthly summary chart.
- API Data Digest — pulls data from a public API daily (e.g. exchange rates) and saves a running log.

Core Resources

- Python for Everybody (Dr. Chuck) — free video course, great for true beginners
- Corey Schafer YouTube channel — clear tutorials on functions, OOP, files, os module
- Real Python (realpython.com) — videos + articles, use for reference when stuck
- freeCodeCamp Python course — good weekend catch-up option
- pandas & openpyxl official docs — for Week 2–3 automation work